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Prediction of CO₂ Solubility in Novel Ionic Liquids Via Transfer Learning-Based Neural Networks

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Abstract

The rational design of high-performance ionic liquids (ILs) for CO₂ capture is hindered by the inability of conventional machine learning methods to predict solubility in novel/untested ILs lacking experimental data. To address this fundamental limitation, we propose a transfer learning-based deep neural network framework for zero-label CO₂ solubility prediction in ILs. Our model innovatively integrates: (1) Shared feature embedding layers that extract domain-invariant representations from both labeled (source) and unlabeled (target) ILs, and (2) A feature-similarity weighted loss function that dynamically aligns source-target feature distributions to enable knowledge transfer. Crucially, this framework requires only molecular structures of target ILs—eliminating experimental solubility measurements. Extensive evaluations on novel IL [N4,1,1,1][Tf2N] demonstrate the proposed framework reduces prediction errors by 4.55–5.02 percentage points in MAE/RMSE over state-of-the-art models (GB, RF, SVR). Ablation studies confirm the hybrid loss function's necessity (2.46–2.56% performance gain) and identify optimal 3-layer encoding depth—outperforming 2-layer (underfitting) and 4-layer (overfitting) variants. This work bridges a critical gap in IL screening pipelines, enabling accelerated discovery of next-generation CO₂ capture solvents.

Key Words: Ionic liquids, CO₂ solubility prediction, Carbon capture, Transfer learning, Deep neural networks

Introduction

The severe challenge of global climate change primarily stems from substantial carbon dioxide (CO₂) emissions resulting from fossil fuel utilization. Increasing concentrations of CO₂ accelerate glacial melting and sea-level rise, threatening to inundate low-lying nations. Furthermore, elevated CO₂ levels directly impact human health (Liao, et al., 2025). Large-scale reduction of CO₂ emissions has become an international consensus, and Carbon Capture, Utilization, and Storage (CCUS) technology represents a critical pathway for achieving deep decarbonization. Within the CCUS chain, efficient and low-cost

CO₂ capture technology is the core component. Post-combustion capture is the most widely deployed technological route, with amine-based solvent absorption being the industrial standard. However, amine-based methods suffer from inherent drawbacks, including high solvent volatility, degradation leading to corrosive byproducts, and high energy consumption during solvent regeneration. Consequently, there is an urgent need to develop greener and more efficient alternative solvents.

Ionic Liquids (ILs), owing to their unique advantages such as low volatility, exceptional thermal/chemical stability, and tunable molecular structure (enabling tailored CO₂ affinity), are regarded as highly promising next-generation materials for CO₂ capture (Liu, et al., 2023). Nevertheless, screening candidates with superior capture performance that also meet economic and engineering requirements from the vast combinatorial space of possible ILs presents a formidable challenge. Traditional trial-and-error experimental approaches are time-consuming, labor-intensive, and costly (Lei, et al., 2014). Machine Learning (ML) methods, including Support Vector Machines (SVM), and their optimized variants (e.g., LSSVM, CSA-LSSVM, SVM-GC), have achieved significant progress in predicting the CO₂ solubility of known ILs (Fu, et al., 2025; Laakso, et al., 2025; Liu, et al., 2023; Rahimi, et al., 2025; Sun, et al., 2023; Yang, et al., 2024). With the development of technology, some deep learning methods have been used to predict the properties of ionic solutions (Rahimi, et al., 2025). For instance, by dividing ionic liquids into dozens of ionic fragments and combining them with artificial neural networks (ANN), a prediction model has been constructed to predict the solubility of CO₂ and nitrogen in ionic liquids (Tian, et al., 2023). A method based on message-passing neural networks learns the molecular structure information of ionic solutions through neural networks and uses it to predict the CO₂ solubility of known ILs (Jun, et al., 2025). However, existing methods share a fundamental limitation: they primarily rely on modeling ILs with known solubility data and are incapable of effectively predicting the CO₂ solubility of novel/untested ILs for which no experimental measurements exist (Ali, et al., 2017; Kamgar and Rahimpour, 2016). This critical bottleneck severely hinders the rational design and high-throughput screening of ILs.

Addressing this core challenge, this study proposes an innovative solution: the development of the first transfer learning (TL)-based deep learning framework designed to tackle the problem of predicting CO₂ solubility for novel ILs under the scenario where the target domain is completely unlabeled (i.e., possesses no known CO₂ solubility data). Deep learning (DL), characterized by its ability to automatically learn hierarchical feature representations from complex, high-dimensional data through multi-layered artificial neural networks, has demonstrated remarkable success in diverse scientific domains, including chemical property prediction (Jing, et al., 2019; Jing, et al., 2021; Jing, et al., 2022). Its power lies in capturing intricate, non-linear relationships within the data that are often intractable for traditional methods. However, DL models typically require large amounts of labeled data for training, which is precisely the limiting factor when dealing with novel ILs. Transfer learning (TL) emerges as a powerful paradigm to mitigate this data scarcity challenge (Jing, et al., 2020). TL leverages knowledge (e.g., learned features, model parameters) acquired from a source domain (typically with abundant labeled data) to improve learning and prediction in a different but related target domain (with limited or no labeled data). The core hypothesis of TL is that domains share underlying commonalities, allowing knowledge transfer to be beneficial. The fundamental motivation stems from the hypothesis that structurally similar ILs should exhibit similar CO₂ dissolution behavior. Building upon the capabilities of DL for feature extraction and the principles of TL for domain adaptation, we constructed a deep neural network model, with its core innovations residing in: (1) Shared Feature Embedding Layers: Shared network layers are designed to learn latent common feature representations for both the source domain (labeled, known ILs) and the target domain (unlabeled, novel ILs). This architecture embodies a key TL strategy, forcing the model to discover a feature space where representations from both domains are aligned, facilitating potential knowledge transfer. (2) Feature Similarity-Based Weighted Loss Function: A novel weighted loss term is proposed, which is based on (Xu, et al., 2019). This loss not only considers the solubility prediction error on the source domain data but,

more importantly, dynamically adjusts network parameters by leveraging the similarity between source and target domain samples within the shared feature space. This mechanism directly operationalizes the TL hypothesis for novel IL prediction: it aims to guide the model to learn features from the source domain that are highly relevant to the target domain task, thereby facilitating effective knowledge transfer from known ILs to novel ILs. By prioritizing source samples that are ‘closer’ to the target samples in the learned feature space, the model focuses its learning on the most transferable knowledge. The optimization objective of this framework can be decomposed into efficiently solvable convex problems. Compared to existing work, the key breakthroughs of this study are:

Pioneering Transfer Regression Framework: It represents the first application of the transfer learning paradigm to CO₂ solubility prediction in ILs, filling the research gap in zero-label prediction for novel/unknown ILs.

Prediction Without Experiments: It enables the prediction of novel IL performance solely utilizing labeled source domain IL data and the structural information of unlabeled target domain ILs (requiring no experimental solubility data for the target domain), significantly reducing screening costs.

Deep Feature Learning and Transfer Mechanism: Through the design of shared layers and the weighted loss function, it achieves deep mining and adaptive transfer of common knowledge across domains.

This work makes three key contributions: (1) It is the first to propose and construct a transfer regression deep learning framework specifically for the challenging task of zero-label CO₂ solubility prediction in novel ILs. (2) It designs a novel network architecture incorporating shared feature layers and a feature similarity-based weighted loss function to address the critical issue of cross-domain knowledge transfer. (3) It conducts comprehensive experimental validation on multiple benchmark datasets for novel ILs, robustly demonstrating the effectiveness of the proposed method and the crucial role of its core components. This work provides a powerful computational tool to accelerate the discovery and design of high-performance, low-cost CO₂-capturing ILs.

Results

Prediction Performance for CO₂ Solubility in Novel ILs

Table 1—Performance of different regression prediction methods on [N4,1,1,1][Tf2N], [N2,1,1,3][Tf2N], and [BMIM][TDfO], respectively. The results in the table are the average values of three separate runs. Gradient Boosting - GB, Random Forest - RF, Support Vector Regression - SVR.

Test IL	Methods	MSE	RMSE	MAE
[N4,1,1,1][Tf2N]	GB	0.0096	0.0958	0.0818
	RF	0.0093	0.0960	0.0905
	SVR	0.3649	0.6041	0.5960
	Ours	0.0023	0.0456	0.0363
[N2,1,1,3][Tf2N]	GB	0.005	0.0689	0.0635
	RF	0.0056	0.0745	0.0718
	SVR	0.4097	0.6401	0.6385
	Ours	0.0022	0.0462	0.0336
[BMIM][TDfO]	GB	0.0192	0.1386	0.1138
	RF	0.024	0.1548	0.1282
	SVR	0.0567	0.2381	0.2134
	Ours	0.0120	0.1084	0.0887

The experimental results for the three test ILs—[N4,1,1,1][Tf2N], [N2,1,1,3][Tf2N], [BMIM][TDfO], are presented in Table 1. The values reported in the tables represent the average performance metrics obtained over three independent runs for each model. During the training of all four models, no label information from the test sets was utilized. Furthermore, hyper-parameters were meticulously tuned via grid search to ensure each model achieved its optimal performance.

As clearly demonstrated by the results, our proposed method consistently and significantly outperformed all existing popular regression prediction methods across all experiments. This superiority is evident in the three widely used regression evaluation metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). For instance, on the [N_{4,1,1,1}][Tf₂N] test set (Table 1), our model achieved MSE, RMSE, and MAE values of 0.0023, 0.0456, and 0.0363, respectively. This represents an reduction of 0.7, 5.02, and 4.55 percentage points over the best results obtained by the benchmark methods. Analysis of Table 1 reveals several noteworthy observations: (1) Ensemble Advantage: Ensemble machine learning algorithms (Gradient Boosting - GB, Random Forest - RF) consistently outperformed the single-model algorithm (Support Vector Regression - SVR). This suggests that ensemble methods are more suitable for tackling the challenging problem of transfer regression prediction. (2) Significant Superiority of Proposed Method: The results from the popular benchmark regression methods (GB, RF, SVR) were markedly inferior to those achieved by our proposed approach. This observation highlights two critical points: Existing methods, constrained to training solely on labeled source domain data, are inherently incapable of accurately predicting the CO₂ solubility of novel/unseen ILs (target domain). The encouraging performance gap strongly validates the effectiveness of our novel method. By leveraging the feature-sharing layers and the specially designed optimization function, our framework successfully transfers valuable knowledge from the source domain to the target domain, enabling accurate solubility prediction for novel ionic liquids.

Ablation Study on the Proposed Hybrid Loss Function

To evaluate the effectiveness of the proposed hybrid loss function, we conducted an ablation study. Specifically, we removed the feature norm adaptive loss and the feature norm similarity loss components designed for the source and target domains. This modification reduces the model to a conventional neural network, denoted as Ours w/o TL. As shown in Table 2, comparative results across the three test ILs consistently demonstrate that the model employing the hybrid loss function outperforms its counterpart without these components (Ours w/o TL). For instance, on the test set for the ionic liquid [N_{4,1,1,1}][Tf₂N], the model with the hybrid loss achieved superior performance across all three evaluation metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). Specifically, it yielded reduction of 0.3%, 2.56%, and 2.46% in MSE, RMSE, and MAE, respectively. These results demonstrate that the hybrid loss function is both effective and essential for enabling successful transfer regression prediction.

Table 2—Performance comparison of models with and without the hybrid loss function on [N4,1,1,1][Tf2N], [N2,1,1,3][Tf2N] and [BMIM][TDfO], respectively. The results in the table are the average values of three separate runs.

Test IL	Methods	MSE	RMSE	MAE
[N4,1,1,1][Tf2N]	Ours w/o TL	0.0053	0.0712	0.0627
	Ours	0.0023	0.0456	0.0363
[N2,1,1,3][Tf2N]	Ours w/o TL	0.0030	0.0513	0.0409
	Ours	0.0022	0.0462	0.0336
[BMIM][TDfO]	Ours w/o TL	0.0165	0.1264	0.1041
	Ours	0.0120	0.1084	0.0887

Impact of Feature Encoding Network Depth on Predictive Performance

The impact of feature encoding network depth within the proposed methods was rigorously evaluated across three ionic liquid data sets, as summarized in Table 3. The proposed architecture (3-layer) consistently outperformed both shallower (2-layer) and deeper (4-layer) variants in all metrics — MSE, RMSE, and MAE — demonstrating its superior capability in capturing complex structure-solubility relationships. The proposed method achieved a 0.22% reduction in MSE compared to the 4-layer model (from 0.0045 to 0.0023), and a 0.17% reduction compared to the 2-layer model (from 0.0040 to 0.0023). This significant error reduction highlights the limitations of both under-parameterized (2L) and over-parameterized (4L) networks.

Table 3—Performance comparison of varied network depths for feature encoding layers on [N4,1,1,1][Tf2N], [N2,1,1,3][Tf2N] and [BMIM][TDfO], respectively. The results in the table are the average values of three separate runs.

Test IL	Methods	MSE	RMSE	MAE
[N4,1,1,1][Tf2N]	Four layers	0.0045	0.0655	0.0561
	Two layers	0.0040	0.0625	0.0534
	Ours	0.0023	0.0456	0.0363
[N2,1,1,3][Tf2N]	Four layers	0.0046	0.0673	0.0474
	Two layers	0.0030	0.0530	0.0444
	Ours	0.0022	0.0462	0.0336
[BMIM][TDfO]	Four layers	0.0186	0.1318	0.1120
	Two layers	0.0215	0.1438	0.1183
	Ours	0.0120	0.1084	0.0887

The 2-layer network's limitations in modeling nonlinear interactions are evident in its degraded performance on test data. The 4-layer network's decline across all ILs suggests overfitting due to excessive parameters. These results confirm that the depth of feature encoding network is a critical hyper-parameter. The proposed 3-layer design delivers statistically significant and practically meaningful gains over alternative depths, establishing its efficacy for robust CO₂ solubility prediction in novel ionic liquids.

Conclusions

This study successfully addresses the critical challenge of predicting CO₂ solubility in novel ionic liquids (ILs) lacking experimental data by pioneering a transfer learning (TL)-based deep learning framework. The core innovation lies in the development of a deep neural network featuring shared feature embedding layers and a novel feature similarity-based weighted loss function. This architecture enables the effective transfer of knowledge from labeled source domain ILs to completely unlabeled target domain novel ILs. Comprehensive experimental validation across multiple novel ILs demonstrated that the proposed method significantly outperforms state-of-the-art regression models (including Gradient Boosting, Random Forest, and Support Vector Regression) across key metrics (MSE, RMSE, MAE). Ablation studies confirmed the essential contribution of the hybrid loss function to enabling successful transfer, while investigations into network depth established the optimality of the proposed 3-layer feature encoder for capturing complex structure-solubility relationships. This work provides the first computational tool capable of zero-label prediction for IL CO₂ solubility, marking a substantial advancement in accelerating the rational design and high-throughput screening of next-generation CO₂ capture solvents.

Discussion

The results underscore a fundamental limitation of conventional machine learning approaches, including powerful ensemble methods: their inability to generalize effectively to novel ILs when trained solely on existing labeled data. The significant performance gap between our TL framework and these benchmarks highlights the success of explicitly modeling and transferring domain-invariant knowledge through shared feature representations and similarity-driven optimization. The effectiveness of the feature similarity-based weighted loss function, as validated by the ablation study, demonstrates that prioritizing source domain samples structurally similar to the target ILs within the learned feature space is crucial for effective knowledge transfer. This mechanism operationalizes the hypothesis that structurally analogous ILs exhibit correlated solvation behavior. Furthermore, the finding that a 3-layer network depth yielded optimal performance, outperforming both shallower and deeper architectures, indicates that capturing the intricate nonlinear relationships between IL structure and CO₂ solubility requires a carefully balanced level of model complexity – sufficient to learn rich representations without succumbing to overfitting.

This framework's capability to predict properties of entirely novel ILs using only their structural information and knowledge from known ILs represents a paradigm shift. It drastically reduces the reliance on costly and time-consuming experimental screening, paving the way for accelerated discovery of high-performance, low-cost ILs for carbon capture. Future work should explore the generalizability of this TL approach to predicting other physicochemical properties of novel materials and investigate integrating additional molecular descriptors or multi-task learning for further enhanced predictive power and mechanistic insight.

While technological innovations like the TL framework presented here are essential for accelerating the discovery of efficient CO₂-capturing ILs, realizing the full potential of Carbon Capture, Utilization, and Storage (CCUS) demands more than scientific breakthroughs. Widespread deployment hinges critically on robust policy frameworks and legal instruments. Effective carbon pricing mechanisms (e.g., carbon taxes, emissions trading systems) are vital to create a tangible economic value for captured CO₂, incentivizing investment in capture technologies like IL-based systems. Complementary policy incentives, such as targeted subsidies, tax credits for CCUS projects, and government-backed loan guarantees, are necessary to bridge the current cost gap and mitigate the financial risks associated with early adoption and scaling. Concurrently, stringent regulatory mandates and emissions standards are required to enforce emission reductions and create a stable market for CCUS solutions. Therefore, the path to deep decarbonization through CCUS necessitates a synergistic approach: continuous advancement in core technologies like the solvent design tool developed here must be paralleled by the establishment of supportive, long-term policies and binding regulatory frameworks that encourage innovation, de-risk investment, and mandate action.

Material and methods

Data collection

This study used a comprehensive database compiled from various reliable sources, including parameters such as temperature, pressure, building groups of ILs and the CO₂ solubility. In total, 3713 data points on the CO₂ solubility measured in 20 different kinds of ILs are collected. The temperature range of the data set is 273.15 ~ 453.15 K, the pressure range is 0.00826 ~ 499.9 bar, and the mole fraction of CO₂ solubility range is 0.0000693 ~ 0.9516. These data are derived from an earlier study (Song, et al., 2020), with updates applied. Table 1 summarizes the ILs studied for CO₂ solubility, detailing their cation/anion names, the temperature, pressure, and CO₂ solubility ranges, as well as the number of data points contained in each IL. In order to achieve an accurate description of diverse ILs, ILs are analyzed by decomposing them into building groups. This decomposition is based on three categories: the cation cores, cation substituents, and anions. The cations include imidazole, pyridine, methyl, phosphorus, etc., and anions contain hexafluorophosphate,

bis (trifluoromethylsulfonyl) imide, trifluoromethylsulfonate, perfluorooctane sulfonate, diethyl phosphate, etc. Additionally, detailed CO₂ solubility data, as well as the full list of building groups showing their numbers of occurrence in each IL, are provided in the Supporting Information.

Table 4—Key parameters of the ionic liquids investigated in this study.

No.	IL	T (K)	P (bar)	CO ₂ (mole fraction)	No. of data points
1	[BMIM][NfO]	293.1500 - 343.1500	12.2100 - 82.8700	0.1373 - 0.9000	71
2	[BMIM][TfO]	298.2000 - 323.2000	19.1000 - 141.1000	0.1043 - 0.9516	59
3	[BMIM][Tf2N]	273.1500 - 453.1500	0.0102 - 499.9000	0.0001 - 0.8032	681
4	[BMIM][TfO]	294.8500 - 344.5500	0.4835 - 375.0000	0.0089 - 0.6720	131
5	[BMPYR][Tf2N]	283.1500 - 413.2000	0.0124 - 484.0000	0.0001 - 0.8530	174
6	[C3MPYR][Tf2N]	298.1500 - 373.1500	5.2000 - 471.0000	0.0930 - 0.7870	83
7	[C5MPYR][Tf2N]	298.1500 - 373.1500	0.0970 - 484.2000	0.0020 - 0.7850	160
8	[C6F9MIM][Tf2N]	298.1300 - 333.3000	0.0157 - 92.3000	0.0002 - 0.8530	59
9	[DMIM][Tf2N]	298.1500 - 343.1500	1.0000 - 201.5000	0.0260 - 0.8780	81
10	[EMIM][DEPO4]	298.4000 - 333.3000	0.2379 - 71.2000	0.0031 - 0.6500	38
11	[EMIM][Tf2N]	283.4300 - 453.1500	0.1000 - 478.5000	0.0040 - 0.7820	568
12	[EMIM][TfO]	303.1500 - 353.1500	1.3620 - 378.0000	0.0139 - 0.6268	121
13	[HMIM][Tf2N]	273.1500 - 413.2000	0.0083 - 452.8000	0.0001 - 0.8333	652
14	[N2,1,1,3][Tf2N]	313.1500 - 333.1500	0.0980 - 94.6600	0.0018 - 0.7390	47
15	[N4,1,1,1][Tf2N]	282.9300 - 343.0700	0.0600 - 203.7000	0.0012 - 0.8790	43
16	[OMIM][PF6]	303.1500 - 353.1500	1.2870 - 92.8800	0.0231 - 0.7550	62
17	[OMIM][Tf2N]	297.5500 - 353.1500	1.0000 - 348.0000	0.0250 - 0.8456	238
18	[P6,6,6,14][C12PhSO3]	303.1500 - 363.1500	3.9000 - 113.7000	0.0971 - 0.7435	112
19	[P6,6,6,14][Cl]	302.5500 - 363.6800	1.6800 - 245.7000	0.1190 - 0.8240	76
20	[P6,6,6,14][Tf2N]	292.8800 - 375.3500	0.4846 - 493.5000	0.0115 - 0.8790	257

Data preprocessing

To ensure model stability and result reliability, it's crucial to handle outliers that may stem from experimental errors or input mistakes. Hence, during data preprocessing, we utilize the EllipticEnvelope model from Python's Scikit-learn library for the abnormal data points detection. Subsequently, we apply the Min-Max normalization method to eliminate the influence of different scales among the features. It will scale the data between 0 and 1. The formula is as follows:

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min}) \quad (1)$$

Here, x' denotes the normalized data value, x denotes the original value to be normalized, $x_{\max}$ denotes the maximum value of the given attribute, and $x_{\min}$ refers to the minimum value of the given attribute.

The input features considered in this study are limited to experimental conditions, such as temperature and pressure, as well as the properties of the ionic liquid solution, specifically the constituent groups of the ionic liquids. The selection of these inputs is advantageous as they are generally more readily accessible.

Our objective is to develop a predictive model based on experimentally measured data, enabling accurate estimation of the carbon dioxide solubility in novel/unseen ionic solutions that have not been experimentally characterized. To evaluate our method for predicting the solubility of novel ionic solutions, we established a transfer learning regression benchmark. This benchmark served as the testing ground for our approach.

The methodology for constructing this benchmark is detailed as follows: The experimental data of CO₂ solubility of 20 different ionic solutions were divided into a training set (16 ILs), validation set (1 ILs) and 3 independent test sets. This partitioning was essential for developing predictive models and subsequently evaluating their ability to forecast CO₂ solubility in new/unseen IL. The training set comprised data from the 16 ionic solutions featuring the densest experimental datasets, ensuring the model learned robust patterns related to CO₂ solubility. The validation set is used to select the hyper-parameters of the model. The remaining three ionic solutions—[N4,1,1,1][Tf2N], [N2,1,1,3][Tf2N], and [BMIM][TDfO]—were reserved as four independent test sets to evaluate the prediction effect of the trained model on new/unseen ionic solutions. It is worth emphasizing that during the training phase of our model, the label information of the test set was not utilized, thereby precluding any potential issue of data leakage.

Method

As illustrated in Fig. 1, our framework comprises a shared feature encoding network F and a regression prediction network G . The feature extraction module F consists of three fully connected layers, three dropout layers, and three normalization layers. G represents the regression predictor with one fully connected layers.

We denote the parameters of G and F by θ_g and θ_f , respectively. Given source domain data $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ of n_s labeled samples and target domain data $D_t = \{x_i^t\}_{i=1}^{n_t}$ of n_t unlabeled samples. Our objective is to develop an algorithm that computes transferable domain features $f = G(F(\cdot))$ using only supervision from the source domain. This study hypothesizes that ionic solutions with greater similarity should exhibit correspondingly similar CO₂ solubility properties. Consequently, aligning the feature distributions of the source and target domains within the feature space is expected to be beneficial for the domain adaptation prediction task.

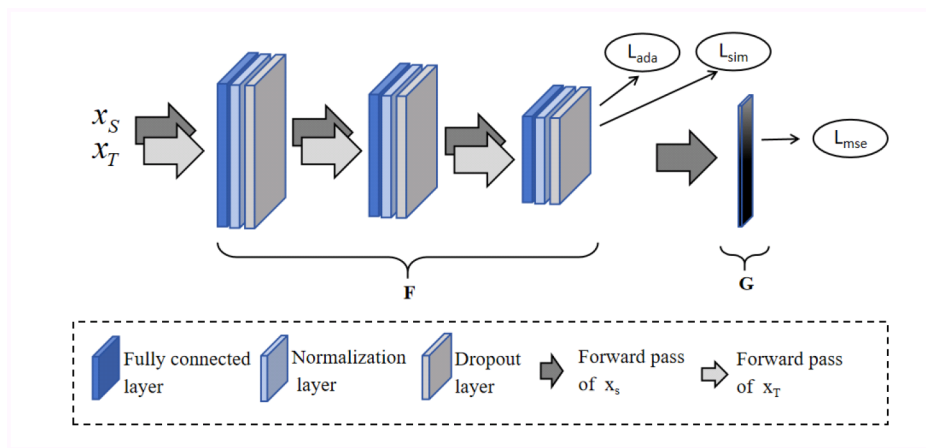


Figure 1—The overall framework of our proposed transfer regression prediction method. The backbone network F represents the shared feature encoding network. G represents the regression prediction layer. During each iteration, we adaptively enlarge the feature norms of each sample in both the source and target domains by a step size Δd , as moving away from small-norms regions facilitates feature transfer. Concurrently, we compute feature similarity between domains and align their deep feature representations. This alignment enhances the statistical similarity between domains, thereby reducing the domain discrepancy to improve model performance on the target domain. The final optimization objective combines the regression prediction loss with these two aforementioned losses.

To achieve this, we propose a hybrid loss function comprising three key components: (1) the source domain regression loss, (2) a feature norm adaptation loss for both source and target domains, and (3) a feature norm similarity loss between the source and target domains. By iteratively refining the alignment of feature norms between the domains, this approach facilitates the effective prediction of CO₂ solubility for novel/unknown ionic solutions. Specifically, We employ the mean squared error (MSE) loss to progressively guide the model in learning the quantitative relationship between data features and CO₂ solubility.

$$L_{mse} = \frac{1}{n} \sum_{i=1}^n (\hat{y} - y_i)^2 \quad (2)$$

Through adaptive feature norm adjustment, the feature norm adaptation loss function computes a specific loss term based on the L2 norm (Euclidean norm) of feature embeddings. It encourages the model to progressively prioritize feature embeddings with larger norms.

$$L_{ada} = \frac{1}{n} \sum_{i=1}^n \left(\|F(x_i, \theta_f)\|_2 + \Delta d - \|F(x_i, \hat{\theta}_f)\|_2 \right) \quad (3)$$

Here, $\|\cdot\|_2$ represents the L2 norm, x_i represents the i -th sample. θ_f and $\hat{\theta}_f$ represent the updated and updating parameters in the previous and current iterations respectively. Δd is a positive residual scalar that controls the expansion of the feature norms. In each iteration, the feature norms are encouraged to expand by a step size of Δd relative to individual examples, based on their feature norms computed from the previous model parameters in the last iteration.

Additionally, we introduce a feature norm similarity loss between source and target domains to progressively align their feature representations.

$$L_{sim} = \frac{1}{n^2} \|Cov(S) - Cov(T)\|_F^2 \quad (4)$$

Here, $Cov(S)$ and $Cov(T)$ represent the covariance matrices of the source and target domain data, respectively. The squared Frobenius norm $\|\cdot\|_F^2$ quantifies the discrepancy between these two covariance matrices. The term n denotes the dimension of the covariance matrices, and division by n^2 serves as a normalization factor to mitigate the influence of dimensional differences on the loss value. By minimizing the divergence between the source and target domain covariance matrices, we aim to align the feature distributions across domains, thereby enhancing the model's generalization capability on the target domain.

Ultimately, the proposed hybrid loss function takes the following form:

$$Loss = L_{mse} + \lambda_1 \cdot L_{ada} + \lambda_2 \cdot L_{sim} \quad (5)$$

Here, λ_1, λ_2 are the hyper-parameters that control the weights of the two types of losses respectively.

Baselines

Given that existing machine learning methods for CO₂ solubility prediction primarily focus on modeling labeled data, while lacking effective models for predicting the solubility of CO₂ in new/unseen ionic solutions, we compare GIFT method against several widely-used regression prediction methods. These methods include Support Vector Regression (SVR), Random Forest Regression (RFR), and Gradient Boosting Regression (GBR) Algorithms. The experimental procedure involved training the models using the training set, optimizing hyper-parameters via the validation set, and finally evaluating the model performance on four independent test set. To ensure fairer comparisons among different methods, we configured multiple sets of hyper-parameters for each of the three benchmark methods and employed grid search to select the optimal hyper-parameters. To ensure robustness, we calculated and reported the average evaluation metrics for each task based on three random runs.

Evaluation metrics

Evaluating the performance of regression method is crucial for understanding how well they predict continuous outcomes. Three most commonly used metrics for assessing regression model performance are used in this paper: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE). MAE measures the average absolute difference between the actual values and the predicted values. It provides a straightforward interpretation of the average error magnitude. MSE measures the average of the squared differences between actual and predicted values. Squaring the errors gives higher weight to

larger errors. RMSE is the square root of MSE and provides an error metric in the same units as the target variable, making it more interpretable.

All experiments were conducted using Python, leveraging the scikit-learn library and the PyTorch framework, on a server running Red Hat Enterprise Linux 8.1 (Ootpa) equipped with an NVIDIA V100s GPU (32 GB).

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